



JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)

WEEK 3: ANGLES OF ELEVATION AND DEPRESSION

TOPIC:

Definition and Applications; Solving Problems Using Scale Drawings

A. ANGLE OF ELEVATION

Definition

The **angle of elevation** is the angle between the horizontal line and the line of sight when looking **UP** at an object.

Key Features of angles of elevation

- Measured from the **horizontal upward**
- Range: 0° to 90°
- Observer looks **upward**
- Forms a **right-angled triangle**

Triangle Representation

- Opposite side $\rightarrow$ Height
- Adjacent side $\rightarrow$ Distance
- Hypotenuse $\rightarrow$ Line of sight



Example 1

A person stands **50 m** from a building and looks up at **30°**.

- Angle of elevation = 30°
- Distance = 50 m
- Height = unknown

Real-Life Examples of angles of elevation

- Looking at the top of a building
- Watching an airplane
- Observing a bird on a tree
- Viewing a mountain peak
- Looking at a flagpole



B. ANGLE OF DEPRESSION

Definition

The **angle of depression** is the angle between the horizontal line and the line of sight when looking **DOWN** at an object.

Key Features



- Measured from the **horizontal downward**
 - Range: **0° to 90°**
 - Observer looks **downward**
 - Equal to angle of elevation from the other point
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Example 2

From a **40 m** high tower, a person looks down at a car at **25°** .

- Angle of depression = 25°
 - Height = 40 m
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Real-Life Examples

- Looking down from a building
 - Viewing ground from an airplane
 - Standing on a bridge
 - Looking from a hill
 - Watching from a lighthouse
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C. RELATIONSHIP BETWEEN ANGLES

Key Principle



Angle of elevation = Angle of depression

(They are **alternate angles** on parallel lines)

Example 3

If angle of elevation = 40° , then

Angle of depression = 40°

D. RIGHT-ANGLED TRIANGLE COMPONENTS

- **Opposite** → Height
- **Adjacent** → Horizontal distance
- **Hypotenuse** → Line of sight

Trigonometric Ratios

(Introduction): $\tan \theta = \text{Opposite} / \text{Adjacent} = \text{Height} / \text{Distance}$

$\sin \theta = \text{Opposite} / \text{Hypotenuse}$

$\cos \theta = \text{Adjacent} / \text{Hypotenuse}$

E. SOLVING USING SCALE DRAWINGS

Steps



1. Understand the problem
2. Draw rough sketch
3. Choose scale
4. Draw accurate diagram
5. Measure required values
6. Convert to actual values

Example 4 (Height)

- Distance = 30 m
- Angle = 35°
- Scale: 1 cm = 5 m

Drawing:

- Horizontal = 6 cm
- Measured height = 4.2 cm

Actual height:
 $= 4.2 \times 5 = \mathbf{21\ m}$



Example 5 (Distance)

- Height = 50 m
- Angle = 28°



- Scale: 1 cm = 10 m

Measured distance = 9.4 cm

Actual distance:
 $= 9.4 \times 10 = \mathbf{94\ m}$

Example 6 (Ladder Problem)

- Ladder length = 10 m
- Angle = 65°
- Scale: 1 cm = 2 m

Measured height = 4.5 cm

Actual height:
 $= 4.5 \times 2 = \mathbf{9\ m}$



F. PRACTICAL APPLICATIONS

1. Measuring Heights

- Trees, buildings

2. Navigation



- Ships and airplanes

3. Construction

- Ramps, roofs

4. Surveying

- Land measurement

G. PROBLEM TYPES

1. Finding Height

- Given: distance + angle
- Find: height

2. Finding Distance

- Given: height + angle
- Find: distance

3. Finding Angle





- Given: height + distance
- Find: angle

Example 12 (Angle)

- Height = 30 m
- Distance = 50 m

Measured angle $\approx 31^\circ$

H. COMBINED PROBLEMS

Example 13

- Distance between buildings = 80 m
- First building height = 40 m
- Angle = 15°

Second building height ≈ 35 m

Example 14

- Plane height = 5000 m



- Angle = 62°

Horizontal distance $\approx$ **2700 m (2.7 km)**

I. ADVANCED APPLICATIONS

- Changing positions (multiple angles)
- Moving objects (ship, plane)
- Double-triangle problems

J. ACCURACY TIPS

Do This

- Use sharp pencil
- Draw large diagrams
- Label clearly
- Choose correct scale

Avoid

- Wrong angle type
- Mixing opposite/adjacent
- Scale errors
- Poor measurement



CLASSWORK

1. Define angle of elevation
2. Define angle of depression
3. Explain why they are equal
4. Solve height using scale drawing
5. Solve distance from cliff problem
6. List instruments needed
7. Give 3 real-life uses
8. If elevation = 35° , what is depression?
9. Why use scale?
10. What does ladder angle show?

REVISION QUESTIONS

1. Difference between elevation and depression
2. Which is measured upward?
3. Draw elevation diagram
4. Draw depression diagram
5. Identify opposite side
6. Why alternate angles matter
7. Steps in scale drawing
8. Best scale for 100 m
9. Converting drawing to real value
10. Can elevation be 95° ? (No)
11. Formula linking height and distance



12. What is hypotenuse?
13. Why use protractor?
14. Professions that use this
15. Checking accuracy

ASSIGNMENT

Section A: Definitions

- Define and draw:
 - Angle of elevation
 - Angle of depression
- Label diagrams
- Explain equality of angles
- Give 5 real-life examples each



Section B: Heights

Solve using scale drawings:

- Building, tree, tower, kite, pole problems
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Section C: Distances

Find distances using:

- Buildings, lighthouse, plane, bridge
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Section D: Angles

Find:

- Elevation
 - Depression
 - Ladder angle
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Section E: Combined Problems

- Multi-position problems
- Moving observer problems
- Building-to-building problems

